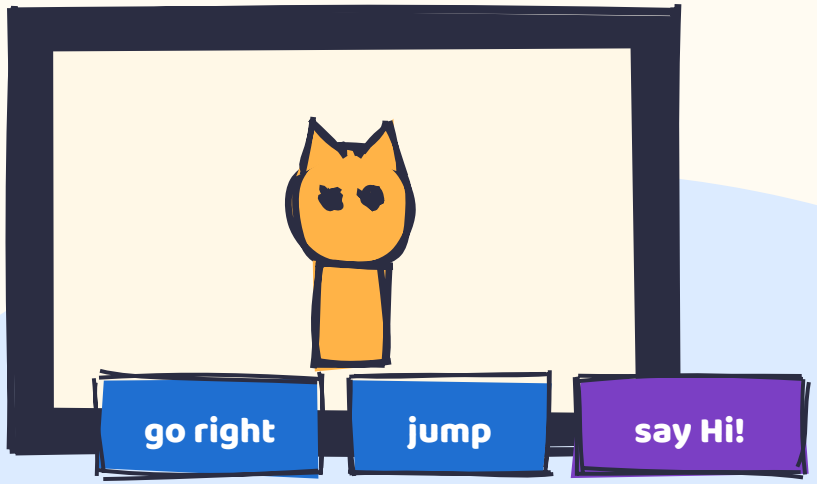
Little Explorers · Class 14

Animated Story App

Make your own cartoon with code!

Sequences | ScratchJr | Debugging



Did you know?
Cartoons are made of many pictures shown very fast. Coders tell characters exactly where to go!

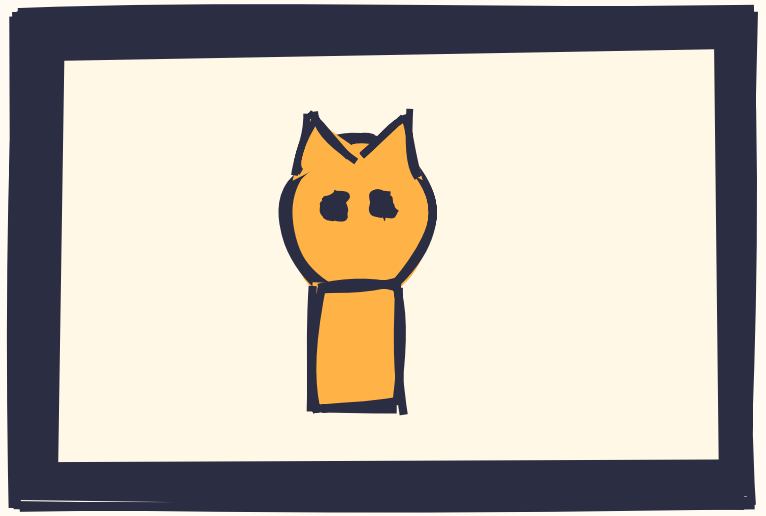
LEARN

Let's wonder...

How do cartoons know what to do?

Who is in charge?

In a cartoon, a cat walks, jumps and talks. Does the cat decide?
No! Somebody gave it **instructions**. Today, **you** are in charge!



The cat thinks?

Magic?

Instructions!

Today: we code a **3-scene story** about a helpful robot in ScratchJr!

Learning objectives

Today I can…

1	**Sequence:** give **step-by-step** instructions in the right order	**Proof:** my partner followed my steps correctly
2	**Code:** make a character **move** and **speak** with ScratchJr blocks	**Proof:** my character moves and says something
3	**Debug:** find and **fix a mistake** in my code	**Proof:** I fixed one bug and can say what it was

LEARN

Objective 2 · Code · video

Video time!



ScratchJr Intro

DevTech Research Group (ScratchJr creators) · about 4 minutes · youtu.be/tEWFDJSmWcw

Watch for...

1. How do you add a **character**?
2. Where are the **blocks** that make it move?
3. How do you start your story?

Scan to watch

After the video: Tell your partner the name of your story's main character. What will it do first?

LEARN

Objective 1 · Sequence

Order matters!

The idea

A **sequence** is a list of steps done **in order**.

If the steps are in the wrong order, things go wrong.
Computers never guess: they just follow the steps.

Quick quiz
Can you put these in order: shoes, socks, walk outside?

Example: brushing your teeth



1, 2, 3!

1. Put toothpaste on the brush. 2. Brush. 3. Rinse. What if we did step 3 first, then step 1? A funny mess! The **order** matters.

Objective 3 · Debug

Bugs are mistakes we can fix

The idea

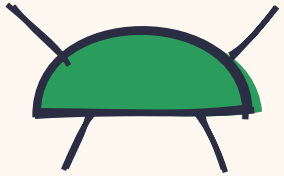
A **bug** is a mistake in our code. Every coder gets bugs, even grown-up experts!

To **debug**: watch carefully, find the block that is wrong, change it, and test again.

Fun fact

The word 'bug' comes from a real moth that got stuck in an old computer!

Example: a recipe mistake



Found it!

If a cake recipe says **salt** instead of **sugar**, the cake tastes funny! We read the recipe, find the mistake and fix it. That is debugging!

Real world

Who codes stories and games?



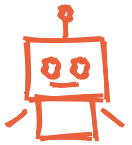
Animators
Make cartoons and films



Game designers
Build games you play on tablets



App makers
Make reading and learning apps



Robot coders
Tell robots what to do, step by step

? What app or game would YOU like to make one day?

Start small
Every big game started as a tiny idea and a few blocks!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Program the teacher!
2. ScratchJr picture blocks
3. Make your character move and speak
4. Plan your 3-scene story
5. Bug hunt!
6. Story challenges

Class game

Program the teacher!

How to play

1. The teacher is a robot that only follows **exact** instructions.
2. Goal: get the robot to the door.
3. Children give one step at a time: **"2 steps forward!"**, **"turn left!"**
4. The robot does EXACTLY what you say, even silly mistakes!
5. Fix the steps until the robot reaches the door.

Then discuss

What went wrong when the robot walked into the wall? How did you fix it?

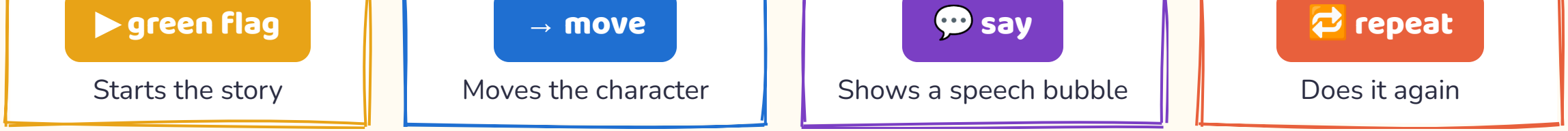


Big idea

Computers do exactly what we tell them, in order. That is called a SEQUENCE.

Objective 2 · Code · meet the blocks

ScratchJr picture blocks



Snap them in a line

Blocks snap together **left to right**. The character does them in that order, one after another.

Tablet tip

Tap a block to test it. Drag it to the coding area to keep it.

MAKE

Objective 2 · Build step 1 of 2

Make your character move and speak

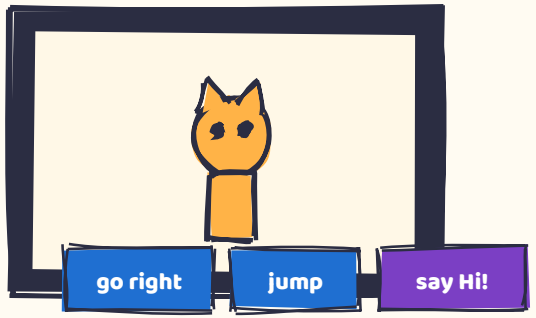
a Tap **+** to add a character. Choose a robot or draw your own!

b Drag the **green flag** block to start.

c Add **move right** 3 times, then a **jump**.

d Add a **say** block and type: **"Hello! I am a helper robot!"**

e Tap the green flag to watch it go!



Check: Your character moves, jumps and speaks when you tap the green flag.

Objective 2 · Code · plan first

Plan your 3-scene story

Scene 1: Problem

draw here

Who needs help? (a lost kitten?)

Scene 2: Help

draw here

How does the robot help?

Scene 3: Happy end

draw here

What happens at the end?

Draw the plan on paper **before** you start coding. Real animators plan with a **storyboard** too!

Debug!

Bug hunt!

Bug	Maybe the problem is…	How we fixed it
The robot walks off the screen	too many move blocks?	_____
Nothing happens	missing green flag?	_____
It talks too fast	add a wait block?	_____

Talk about it

Swap tablets with a partner. Can you find a bug in **their** story and help fix it?

Be kind

Say 'I found a bug, let's fix it together!'

MAKE

Level up!

Story challenges

Choose your challenge



Two characters: make them talk to each other.



Sound effects: record your own voice saying a line.



Repeat dance: make a character dance with a repeat block.



New scene: paint a background for each scene.

Think like an engineer

Why is it a good idea to plan your story before coding?

Boss level

Make your story change scene by itself at the end!

Objective check

Can I really do it? Prove it.

01	Tell 3 steps to brush your teeth in the right order.	○ ○ ○
02	Show your character moving and speaking.	○ ○ ○
03	What bug did you find, and how did you fix it?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Story show: play each group's app on the projector. Clap for every story!

Show what you know

What did we learn?

- [x] A sequence is steps in the right order
- [x] Blocks run left to right, one after another
- [x] Plan the story first, then code it
- [x] Bugs are normal: find them and fix them

At home

Tell your family your robot story. Can they guess the ending?



Exit ticket: what is a bug in coding?

Next class: make a Catch-the-Fruit Game!